

Bakr Matlab

Ottawa, ON | matlab893@gmail.com | (819) 576-7914
bakrmatlab.com | github.com/bakrmatlab | linkedin.com/in/bakr-matlab

Education

CARLETON UNIVERSITY

BSc Computer Science, Cybersecurity stream
Core GPA: 11.0/12.0

Selected study: Algorithm design, data structures and discrete math; operating systems, computer architecture and networks; cryptographic methods and secure software design.

Ottawa, ON

Expected April 2029

ALGONQUIN COLLEGE

Computer Programming Diploma

Selected study: Java and object-oriented design; data structures; relational data and SQL; full-stack web applications developed with a team.

Ottawa, ON

April 2023

Technical Skills

Languages:	TypeScript, JavaScript, Java, Python, SQL
Frameworks:	React, Next.js, React Native, Express, Spring Boot
Backend & Data:	REST APIs, GraphQL, Convex, PostgreSQL, MySQL
Tools:	Git, Linux, Docker, Vercel, JUnit, Postman

Projects

MoneyPal | Personal Finance Web App

2026

github.com/bakrmatlab/MoneyPal | money-pal-wheat.vercel.app

- Developed and deployed a React and TypeScript finance app with a Convex real-time backend and Clerk authentication for wallets, transactions, and spending analytics.
- Implemented simulated user-to-user transfers with recipient validation and wallet checks, updating transaction history and reversing budget spending when transfers are deleted.
- Built monthly budgeting with Convex queries and mutations, scheduled rollovers, overspend warnings, and budget history views.
- Integrated route guards and three-step onboarding to guide new users through wallet and budget setup before dashboard access.

Pet Connect | Social App for Pet Owners

2023

github.com/FinalScript/pet-connect | Team project

- Implemented GraphQL schemas, queries, and mutations for pets and posts in an Express and MySQL backend serving a React Native app.
- Added post lookup, update, and deletion operations plus pet lookup and username validation queries.
- Debugged pet and owner updates and comment and like endpoints during team development.

Manage.sys | Store Management System

2022

github.com/FinalScript/manage.sys | Team project

- Built Spring Boot endpoints for store and employee records backed by PostgreSQL, including update and delete operations.
- Integrated employee edit and delete flows in the React client with the corresponding backend operations.
- Refactored Java store and employee classes as the application expanded.